

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. STA/STB-602: Demand Analysis, Analysis of Income Distribution and Queuing Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Write a note on the equilibrium price. Find the equilibrium price and quantity for crude oil if the demand and supply equations are $Q_d = 500 - 4p$ and $Q_s = 100 + 2p$, respectively. Also sketch a free-hand curve and find the equilibrium price and quantity graphically. [7]
- (b) Define a queuing system and its important components. [7]

Question 2.

- (a) Describe Pigou's method (with assumptions and limitations) for estimating elasticity from both time series data and family budget data. [7]
- (b) Consider a competitive market for which the quantities demanded and supplied (per year) at various prices are given as follows:

Price (Rs.)	600	800	1000	1200
Demand (million)	22	20	18	16
Supply (million)	14	16	18	20

- (i) Calculate the price elasticity of demand when price is Rs. 800.
- (ii) Calculate the price elasticity of supply when price is Rs. 800.
- (iii) What are the equilibrium price and quantity?
- (iv) Suppose the government sets a price ceiling of Rs. 800. Will there be a shortage, and if so, how large will it be?

[7]

Question 3.

- (a) For an exponential income distribution $dF = c e^{-cx} dx$, $c > 0$, $0 < x < \infty$, show that the Lorenz coefficient of concentration is equal to $\frac{1}{2}$. [7]

- (b) The data below relates to the distribution of urban households in India. Fit a Pareto distribution to the data and, with the estimated parameters, derive the Lorenz concentration coefficient and the mean income:

Income Class (Rs.)	Households (%)
Under 500	13.6
500 – 999	28.9
1000 – 1999	32.5
2000 – 2999	10.6
3000 – 3999	5.6
4000 – 4999	3.1
5000 – 5999	1.7
6000 – 7999	1.7
8000 – 9999	0.7
10000 – 14999	0.8
15000 – 24999	0.5
25000 and above	0.3

[7]

Question 4.

- (a) Prove that in a Poisson arrival process, the inter-arrival time distribution follows the exponential distribution. [7]
- (b) Define a machine-servicing model in queuing theory. Derive the expression for the probability of n customers in the system, P_n , for the finite-capacity $(M/M/1) : (N/FCFS)$ model. [7]

Question 5.

- (a) Define the cumulative distribution of the waiting time for a customer who has to wait in the $(M/M/1) : (\infty/FCFS)$ queuing model. [7]
- (b) A one-person barber shop can accommodate a maximum of 5 people at a time (4 waiting, 1 getting a haircut). Customers arrive according to a Poisson distribution with mean rate 5 per hour. The barber cuts hair at an exponential rate of 4 per hour. Find:
- What percentage of time is the barber idle?
 - What fraction of potential customers is turned away?
 - What is the expected number of customers waiting for a haircut?
 - How much time can a customer expect to spend in the barber shop?

[7]

Question 6.

- (a) For the case of s channels with Poisson arrivals at rate λ and exponential service at rate μ , show that the probability that an arrival has to wait is:

$$P(\text{wait}) = \frac{(\lambda/\mu)^s}{s! \left(1 - \frac{\lambda}{s\mu}\right)} \cdot P_0$$

[7]

- (b) At a health centre, one patient arrives every $\frac{1}{2}$ hour. The average treatment time for a patient is 20 minutes. If the hospital has 2 doctors, find:

- (i) Probability that both doctors are idle
- (ii) Probability that exactly one doctor is idle
- (iii) Expected number of patients waiting for a doctor
- (iv) Expected waiting time of a patient

[7]

Question 7. Write short notes on any **three** of the following:

[14]

- (a) Curves of concentration (Lorenz curve)
- (b) Pareto's law of income distribution
- (c) Giffen's paradox
- (d) Transient and steady states in queuing theory